DEPI DATA ANALYSIS TRACK | GRADUATION PROJECT

Road Traffic Accident Analysis

A descriptive Power BI analysis of 12,316 road accident records, showing when accidents are recorded and which driver, vehicle, road, environment and cause conditions are associated with serious and fatal outcomes.

PROJECT TEAM (5 MEMBERS)

Retaj Etman Saber Elgallad Team Leader (TL)

Ziad Hamdy Mostafa Abdelfattah

Habiba Talaat Abd El Azeem Farag

Ibrahim Ahmed Adel Elbyaly

Saif Eslam Fayeq Elsaedy

PROJECT DETAILS

DATASET
Road Traffic Accidents (RTA)

MAIN TECHNOLOGY
Power BI

BUSINESS REQUIREMENTS
40 analytical questions

FINAL DEADLINE
4 December 2026

GITHUB REPOSITORY
https://github.com/retajelgallad/Road-Accidents-Analysis

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1 Executive Summary and Project Overview

1.1 Project overview

Road Traffic Accident Analysis is the graduation project of a five-member team in the DEPI Data Analysis Track. It uses the Road Traffic Accidents (RTA) dataset, which holds **12,316 accident records** described by **32 columns** covering drivers, vehicles, roads, the environment, casualties and reported causes.

The problem. In its raw form the dataset is hard to interpret. Time is stored as text, 16 columns contain missing values, several labels contain typing errors, and 84.6% of accidents are slight injuries, so serious and fatal accidents are easy to overlook.

Why it matters. Serious and fatal outcomes are what road-safety work most needs to understand. Organised evidence on when accidents are recorded, how severe they are and which recorded conditions accompany serious outcomes gives a factual starting point for safety discussions and prioritisation.

Purpose and output. The team will clean and standardise the data in Power Query, answer 40 business questions and present the results in an interactive six-page **Power BI** dashboard. Power BI holds the data model and DAX measures and communicates results through KPI cards, charts and slicers that show counts, percentages and group sizes together. A final report, presentation slides and a short demo video complete the delivery.

Expected value. The findings will be descriptive: a clear view of where accident volume and severity concentrate by hour, weekday, driver, vehicle, road, environment and reported cause, with data limitations kept visible. They will describe associations in historical records and will not prove causes.

1.2 Project at a glance

12,316 accident records	**32** source columns	**40** business questions
6 dashboard pages	**5** team members	**4 December 2026** final deadline

1.3 Intended outputs and document status

- A cleaned, Power BI-ready dataset with a short cleaning log.
- A data dictionary covering all 32 source columns and a data model (schema) diagram.
- An interactive Power BI dashboard with tested DAX measures, filters and KPIs.
- A final report with key findings, limitations and recommendations, presentation slides and a 1–2 minute demo video, with all files organised in the GitHub repository.

> **Document status**
> This document sets out the project plan, requirements and design. The dashboard, findings and recommendations are **planned deliverables**; none is presented here as completed or validated. Figures in Section 3 come from initial profiling of the dataset and will be re-checked on the cleaned data.

2 Problem Statement, Objectives and Scope

2.1 Problem statement

The RTA dataset contains 12,316 accident records and 32 columns on drivers, vehicles, roads, lighting, weather, collision types, causes and severity. As supplied, time is stored as text, 16 columns have missing values (about three in four rows have at least one blank field), some labels contain typing errors, and 84.6% of accidents are slight. It is therefore hard to tell when accidents are recorded, how severity differs between conditions, and which recorded factors are most often linked to serious and fatal outcomes.

2.2 Objectives

Main objective. Clean the RTA data and use it to answer 40 business questions about drivers, vehicles, roads, environment and time, and accident causes and casualties, presenting the results in an interactive Power BI dashboard, a report, slides and a demo video.

Specific analytical objectives

- Clean and standardise the dataset in Power Query while preserving the meaning of source values.
- Measure accident counts, casualties, vehicles involved and the severity distribution.
- Compare severity across driver, vehicle, road, lighting, weather, collision and cause categories.
- Show when accidents are recorded most often by hour of day and day of week.
- Build an interactive six-page Power BI dashboard with filters and summary KPIs.
- Write a report with validated findings, limitations and data-backed recommendations.

2.3 Scope

Included project activities	Out of scope
Data cleaning and preparation in Power Query	Proving the cause of individual accidents
Descriptive analysis and comparisons between categories	Predicting future accidents
Severity and casualty analysis	Maps and geographic analysis (the data has no location)
Hour-of-day and day-of-week patterns	Yearly, seasonal or trend analysis (the data has no date)
Interactive Power BI dashboard, final report, slides and demo video	Speed analysis (the data has no speed field)
Documentation and file organisation on GitHub	Outside data, real-time data feeds, mobile apps or government-system integration

Interpretation principle
The analysis is **descriptive**. It identifies associations between conditions and accident severity in historical records. It does not prove that any condition causes an accident or a more severe outcome, and wording such as “is associated with” is used throughout. Conclusions and recommendations are written only after the data is cleaned and results are validated against it. Advanced statistical tests and predictive models are not part of the required scope.

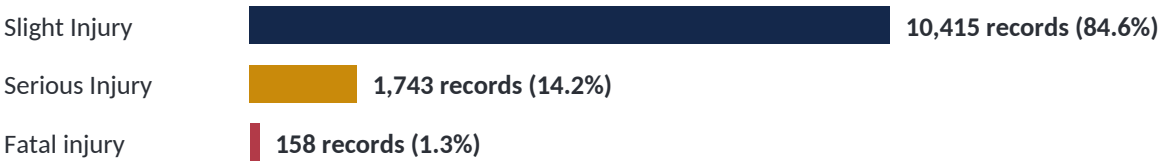
3 Dataset Overview and Data Quality

3.1 Dataset at a glance

The project uses the RTA dataset. Each row is one accident record. The dataset has no calendar date, year, city, GPS coordinates, road name or speed, so time analysis is limited to hour of day and day of week. No outside data is added.

Measure	Value	Note
Accident records (rows)	12,316	One row counted as one accident
Source columns	32	Original names and spellings are kept
Exact duplicate rows	0	
Columns with missing values	16	Listed in Section 3.5
Missing cells	20,057	Sum of blanks over the 16 columns
Rows with at least one blank field	9,427 (76.5%)	
Rows with casualty details	7,873	12,316 minus 4,443 rows with “na”
Rows with “na” in the four casualty fields	4,443 (36.1%)	Class, sex, age band and severity
Total casualties / total vehicles	19,067 / 25,133	Sums of Number_of_casualties and Number_of_vehicles_involved

3.2 Accident severity distribution



Accident severity	Records	Share of records
Slight Injury	10,415	84.6%
Serious Injury	1,743	14.2%
Fatal injury	158	1.3%
Total	**12,316**	**100.0%**

Shares are rounded to one decimal place. Serious Injury plus Fatal injury equals 1,901 records (15.4% of all records). The severity classes are strongly imbalanced, so counts are always shown together with percentages.

3.3 Source columns by category

The 32 source columns fall into six categories: driver, vehicle, road, environment, accident and casualty information. Column names are kept exactly as in the source, including spelling mistakes such as Road_allignment and Area_accident_occured. “Blank” counts are shown where known.

Column	Type	Meaning and notes
DRIVER		
Age_band_of_driver	Text	Under 18, 18–30, 31–50, Over 51 or Unknown.
Sex_of_driver	Text	Male, Female or Unknown.
Educational_level	Text	Driver education level. 741 blank.
Vehicle_driver_relation	Text	Owner, Employee, Other or Unknown. 579 blank.
Driving_experience	Text	Below 1yr to Above 10yr, or No Licence. 829 blank.
VEHICLE		
Type_of_vehicle	Text	17 categories (automobile, lorry, public bus, motorcycle, taxi and others). 950 blank. Some labels contain “?” in place of a dash, for example Lorry (41?100Q).
Owner_of_vehicle	Text	Owner, Governmental, Organization, Other. 482 blank.
Service_year_of_vehicle	Text	Below 1yr to Above 10yr. 3,928 blank and 2,883 Unknown.
Defect_of_vehicle	Text	“No defect” or codes 5 and 7 (not documented). 4,427 blank.
Vehicle_movement	Text	13 categories (going straight, overtaking, U-turn, reversing and others). 308 blank.
ROAD		
Area_accident_occured	Text	14 area types (residential, office, school, market, rural and others). 239 blank. One label is two values joined together.
Lanes_or_Medians	Text	Lane or median layout (one way, divided, undivided and others). 385 blank.
Road_allignment	Text	Road alignment and terrain (tangent, curve, steep grade, mountainous). 142 blank.
Types_of_Junction	Text	No junction, T / Y / O shape, Crossing, Other. 887 blank.
Road_surface_type	Text	Asphalt, Earth, Gravel and others. 172 blank.
Road_surface_conditions	Text	Dry, Wet or damp, Snow, Flood over 3cm.
ENVIRONMENT		
Light_conditions	Text	Daylight, or darkness with lights lit, lights unlit or no lighting.
Weather_conditions	Text	9 categories (for example Normal, Raining, Windy, Cloudy, Snow, Fog or mist, Other).
ACCIDENT		
Time	Time	Time of day, stored as text (H:MM:SS) and converted to a time type. Hours before 10 have no leading zero (2,808 rows).

Column	Type	Meaning and notes
Day_of_week	Text	Monday to Sunday.
Type_of_collision	Text	10 categories (vehicle with vehicle, rollover, with animals, with pedestrians and others). 155 blank.
Number_of_vehicles_involved	Whole number	Vehicles involved, 1 to 7.
Number_of_casualties	Whole number	Casualties, 1 to 8.
Cause_of_accident	Text	Reported cause, 20 categories. Some labels look alike, for example “Overspeed” and “Driving at high speed”.
Accident_severity	Text	Target outcome: Slight Injury, Serious Injury or Fatal injury.
CASUALTY (DESCRIPTION ONLY)		
Casualty_class	Text	Driver or rider, Passenger, Pedestrian, or “na”.
Sex_of_casualty	Text	Male, Female, or “na”.
Age_band_of_casualty	Text	Age group, or “na”. Code “5” appears in 244 rows (not documented).
Casualty_severity	Text	Code 1, 2 or 3 (meaning not documented), or “na”.
Work_of_casuality	Text	Casualty occupation. 3,198 blank.
Fitness_of_casuality	Text	Normal, Deaf, Blind, Other. 2,635 blank. 19 rows read “NormalNormal” (typing error).
Pedestrian_movement	Text	Pedestrian movement, or “Not a Pedestrian”. Some labels contain a typing error (merged text) in place of “stationary”.

Category totals: driver 5, vehicle 5, road 6, environment 2, accident 7 and casualty 7 columns, 32 in total.

3.4 Planned derived columns and analysis definitions

The following helper columns are planned in Power BI. The original columns are not overwritten.

Derived column	Type	Definition
Hour	Whole number	Hour of day (0–23) taken from Time.
Time Band	Text	00–06, 06–10, 10–14, 14–17, 17–20 or 20–24.
Serious or Fatal	Whole number	1 if Accident_severity is Serious Injury or Fatal injury, otherwise 0.
Severity Order	Whole number	Slight = 1, Serious = 2, Fatal = 3; sorts visuals.
Day Order	Whole number	Monday = 1 to Sunday = 7; sorts visuals.
Day Type	Text	Weekend (Saturday and Sunday) or Weekday. The definition is stated wherever a weekend result is shown.
Light Group	Text	Daylight or Darkness (the three darkness values grouped).
At Junction	Text	Yes for any junction type except “No junction”, otherwise No; blanks stay not recorded.

Derived column	Type	Definition
Has Defect	Text	Yes if a defect code is recorded, No if “No defect”, not recorded if blank.

Definitions used in the analysis

- **Accident record:** one row of the dataset, counted as one accident unless the source documentation shows otherwise.
- **Serious-or-fatal share:** Serious Injury plus Fatal injury records divided by all records in the selected filter. Counts and percentages are shown together.
- **Weekend:** Saturday and Sunday (team definition; results change with the definition).
- **Casualty fields:** they describe the recorded casualty. The grain of each field is validated before any casualty count is summed, and casualty columns are used for description only, not as risk factors.

3.5 Data quality profile

Initial profiling found 20,057 missing cells across 16 columns, and 9,427 rows (76.5%) have at least one blank field. The table lists every column with blanks; the counts add up to the 20,057 total.

Column	Blank rows	Share of 12,316 rows
Defect_of_vehicle	4,427	35.9%
Service_year_of_vehicle	3,928	31.9%
Work_of_casuality	3,198	26.0%
Fitness_of_casuality	2,635	21.4%
Type_of_vehicle	950	7.7%
Types_of_Junction	887	7.2%
Driving_experience	829	6.7%
Educational_level	741	6.0%
Vehicle_driver_relation	579	4.7%
Owner_of_vehicle	482	3.9%
Lanes_or_Medians	385	3.1%
Vehicle_movement	308	2.5%
Area_accident_occured	239	1.9%
Road_surface_type	172	1.4%
Type_of_collision	155	1.3%
Road_allignment	142	1.2%
Total missing cells	**20,057**	

Main data quality issues and planned treatment

Issue found	Planned treatment
4,443 rows (36.1%) have “na” in the four casualty fields.	“na” means no casualty detail was recorded. It is kept as its own value.
Time is stored as text; hours before 10 have no leading zero (2,808 rows).	Convert to a time type and derive Hour and Time Band.
Defect_of_vehicle holds undocumented codes 5 and 7 (112 rows).	Keep as recorded, label as “Code 5” and “Code 7”, and do not guess their meaning.
Casualty_severity codes 1–3 and Age_band_of_casualty code “5” (244 rows) are not documented.	Keep as codes and describe the limitation in the report.
Service_year_of_vehicle has 2,883 “Unknown” values on top of 3,928 blanks.	Report both and treat the field with caution.
Typing and encoding errors: merged labels in Pedestrian_movement and Area_accident_occured, “NormalNormal” (19 rows), “?” in vehicle-type labels, stray spaces.	Trim spaces and standardise only labels that clearly mean the same category.
Similar-looking cause labels, for example “Overspeed” and “Driving at high speed”.	Review before any merging; merge only when clearly the same and record the decision.
Only 158 fatal records.	Show group sizes next to rates and avoid strong claims about small groups.

Blank, “na” and “Unknown” are three different things and are never silently merged. A blank is a missing entry, “na” means no casualty detail was recorded, and “Unknown” is a recorded value.

3.6 Cleaning principles

- Keep the original CSV file unchanged and work on a cleaned copy.
- Do not delete a record only because an optional field is missing.
- Do not invent replacement values; keep original column names, including source spellings.
- Record every change, its reason and its effect in the cleaning log.

3.7 Preliminary profiling observations

These first-pass figures came from initial profiling. They are associations, not proven causes, and each will be re-tested on the cleaned data before appearing in the final report. Figures in brackets are group sizes in records.

Topic	Preliminary observation	Caution
Time of day	15:00–18:59 holds 3,979 records (32.3%); the 17:00 hour peaks at 1,228.	Hour only; no date or location.
Lighting	No lighting: 28.1% serious-or-fatal (192) vs 14.9% in daylight (8,798); overall rate 15.4% (1,901 of 12,316).	Small group.
Driver age	Drivers under 18: 21.9% serious-or-fatal (825). Driving experience shows no clear pattern (14–16%).	Descriptive only.

Topic	Preliminary observation	Caution
Vehicles involved	One vehicle: 28.2% serious-or-fatal (1,996) vs 12.8% for two vehicles (8,340).	Descriptive only.
Pedestrian collisions	Fatal rate 2.5% (896) vs 1.2% for vehicle-with-vehicle (8,774).	Only 158 fatal records.
Weekend	Fatal rate 2.3% on Saturday–Sunday vs 0.94% on weekdays.	Depends on definition: with Friday–Saturday the gap is 1.43% vs 1.22%.
Reported causes	"No distancing" is the most frequent cause (2,263); left and right lane changes total 3,281.	Reported, not verified, causes.

4 Project Methodology and Technical Workflow

4.1 Workflow

The project moves from the raw file to a tested dashboard in nine connected steps. Planned activities are listed; none is reported as completed.

Step	Main activities	Primary output
1. Initial dataset inspection	First look in Excel; check missing values, duplicates, wrong labels and data types in Power Query.	Data quality profile
2. Data cleaning and preparation	Trim spaces, fix typing errors, convert Time, add helper columns; keep "na", blanks and "Unknown" distinct.	Cleaned dataset and cleaning log
3. Data dictionary development	Document every source column: meaning, data type and notes.	Data dictionary (32 of 32 columns)
4. Data modelling and relationships	One cleaned Accidents table plus small helper tables for sorting (day of week, severity); check relationships.	Data model / schema diagram
5. DAX measures and KPIs	Build reusable measures (Section 8.3). Each measure is reviewed by a second member.	Reviewed DAX measures
6. Exploratory and descriptive analysis	Compare counts and serious-or-fatal shares across categories; study hour and weekday patterns.	Validated findings
7. Power BI dashboard development	Build six pages with charts, cards, slicers and a data-limitations note.	Interactive dashboard
8. Testing, validation and interpretation	Check key totals and every measure against the cleaned data; test slicers, navigation and interactions; flag groups with fewer than 30 accidents.	Reconciled and tested dashboard
9. Reporting, presentation and delivery	Write the report and recommendations, prepare slides, record the demo video, upload files to GitHub.	Final report, slides, demo video

4.2 Tools and their roles

Tool	Role in the project
Excel	First look at the data, pivot-table reference checks for validation, and the data dictionary workbook.
Power Query	Data cleaning and preparation inside Power BI.
Power BI Desktop	Data model, charts and dashboard.
DAX	Calculations, KPIs and rankings.
GitHub	Project files, documentation, version history and final deliverables.

Tool plan
SQL and Python are not used. Data cleaning and analysis are done with Excel, Power Query, Power BI and DAX. Advanced statistics and predictive models are outside the required scope.

5 Business Requirements and Analytical Questions

The dashboard is required to answer the following 40 questions. Each can be answered from the available columns with a simple DAX measure and a chart, card or table. They are **requirements, not findings**. "Serious or fatal" means Serious Injury or Fatal injury. Questions about casualty attributes account for "na" values and validate the grain of each field before counts are interpreted.

5.1 Driver (Questions 1–8)

No.	Business question
1	Which driver age band is involved in the most accidents?
2	Which driver age band has the highest percentage of serious or fatal accidents?
3	How many accidents involve male drivers and how many involve female drivers?
4	Which driving experience group has the most accidents?
5	Which driving experience group has the highest percentage of serious or fatal accidents?
6	Which education level has the most accidents?
7	How many accidents involve drivers with no licence?
8	What share of accidents comes from each driver-vehicle relation (owner, employee, other)?

5.2 Vehicle (Questions 9–16)

No.	Business question
9	Which vehicle type is involved in the most accidents?
10	Which vehicle type has the highest percentage of serious or fatal accidents (types with at least 30 accidents)?
11	Which vehicle owner type has the most accidents?
12	Which vehicle service-year group has the most accidents?
13	How many accidents involve a vehicle with a recorded defect?
14	Which vehicle movement is the most common in accidents?
15	What is the average number of vehicles involved per accident?
16	What percentage of accidents involve more than two vehicles?

5.3 Road and Junction (Questions 17–24)

No.	Business question
17	Which junction type has the most accidents?
18	What percentage of accidents happens at a junction (any type except “No junction”)?
19	Which lane or median type has the most accidents?
20	Which road alignment has the most accidents?
21	Which road surface type has the most accidents?
22	Which area type has the most accidents?
23	Which type of collision is the most common?
24	Which type of collision has the highest percentage of serious or fatal accidents?

5.4 Environment and Time (Questions 25–32)

No.	Business question
25	At which hour of the day do the most accidents happen?
26	Which day of the week has the most accidents?
27	Which day of the week has the highest percentage of serious or fatal accidents?
28	What percentage of accidents happens in daylight and what percentage in darkness?
29	Which light condition has the highest percentage of serious or fatal accidents?
30	Which weather condition has the most accidents?
31	How many accidents happen on each road surface condition (dry, wet, snow, flood)?
32	What percentage of accidents happens on weekends (Saturday and Sunday)?

5.5 Causes and Casualties (Questions 33–40)

No.	Business question
33	Which cause of accident is the most frequent?
34	Which five causes have the highest percentage of serious or fatal accidents (causes with at least 30 accidents)?
35	What share of all accidents does each of the top five causes represent?
36	What is the average number of casualties per accident?
37	What percentage of accidents has more than one casualty?
38	What share of casualties are drivers or riders, passengers and pedestrians?
39	Which casualty age band has the most casualties?
40	How many accidents are Slight, Serious and Fatal, and what is the percentage of each?

6 Team Members and Project Responsibilities

The project is carried out by five team members. Retaj Etman Saber Elgallad is the Team Leader. The responsibilities below are the primary allocation for the project plan; each member remains responsible for finishing their own tasks.

6.1 Member roles

Retaj Etman Saber Elgallad — **Team Leader (TL)**

- Coordinate the project, monitor deadlines and track tasks and weekly progress.
- Create, manage and organise the GitHub repository and verify that all deliverables are uploaded.
- Oversee the project proposal, risk planning and overall documentation, including the final report.
- Contribute to dataset cleaning and create about half of the DAX measures.

Project stages: Active in every phase; leads planning (Phase 1) and final delivery (Phase 4).

Ibrahim Ahmed Adel Elbyaly — **Data Cleaning and Validation**

- Inspect, clean and prepare the dataset using Excel and Power Query.
- Maintain the data-cleaning log.
- Help validate the cleaned dataset.
- Support report preparation and organisation.

Project stages: Cleaning plan (Phase 2), data preparation (Phase 3), validation and report (Phase 4).

Saif Eslam Fayeq Elsaedy — **Data Modeling and Data Dictionary**

- Develop the data schema and the Power BI data model.
- Document source columns, meanings, data types and relationships in the data dictionary.
- Check that model relationships work correctly.
- Create the remaining DAX measures, following the agreed division of work with Retaj.

Project stages: Dictionary and model (Phase 2), DAX measures (Phase 3).

Habiba Talaat Abd El Azeem Farag — **Dashboard Design and Implementation**

- Plan the dashboard layout and visual hierarchy.
- Select appropriate charts, KPIs and filters.
- Build and organise the dashboard pages in Power BI.
- Review readability, consistency and usability.

Project stages: Dashboard sketches (Phase 2), dashboard build (Phase 3), usability review (Phase 4).

Ziad Hamdy Mostafa Abdelfattah — **Testing, Reporting and Presentation**

- Support validation of calculations and dashboard totals.
- Help prepare the final report.
- Prepare and organise the final presentation slides.
- Communicate validated findings and recommendations.

Project stages: Validation (Phases 3–4), report, slides and presentation (Phase 4).

6.2 Shared responsibilities and handoffs

- All members review the 40 business questions and confirm that the dataset has the needed columns.
- All members test the dashboard, check that totals match the cleaned data and contribute to quality checks.
- The Team Leader checks progress through a weekly progress check; every member is responsible for finishing their own tasks.
- DAX measures are shared between Retaj and Saif, who review each other's results; every measure is checked by a second team member.
- The final report is coordinated by Retaj with Ibrahim and Ziad. Ziad leads the slides. The demo video is recorded by one member with help from the team; the member is to be confirmed.

6.3 Responsibilities by project stage

Project stage (phase)	Lead	Support
Planning, roles, risk plan and questions (Phase 1)	Retaj	All members
Cleaning plan and data dictionary (Phase 2)	Ibrahim, Saif	Retaj
Data model and schema (Phase 2)	Saif	Retaj, Ibrahim
Dashboard layout and sketches (Phase 2)	Habiba	All members review
Data cleaning and cleaning log (Phase 3)	Ibrahim	Retaj
DAX measures (Phase 3)	Retaj and Saif	Ziad, whole team
Dashboard build (Phase 3)	Habiba	Saif, whole team
Totals validation and dashboard testing (Phases 3–4)	Ziad	Ibrahim and all members
Final report (Phase 4)	Retaj (coordination)	Ibrahim, Ziad
Slides and presentation (Phase 4)	Ziad	Whole team
Demo video (Phase 4)	To be confirmed	Whole team
GitHub organisation and upload check (all phases)	Retaj	All members upload own files

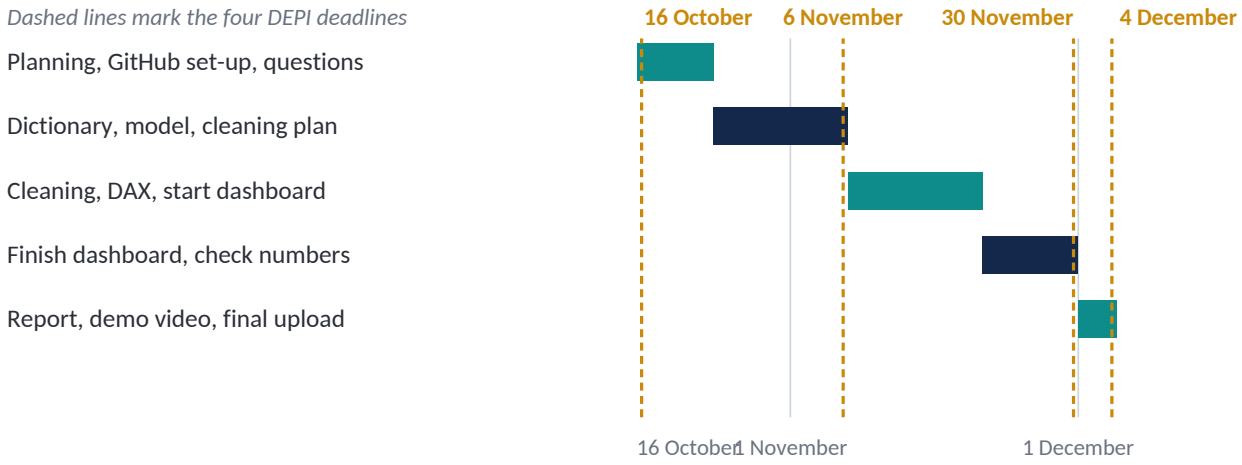
7 Project Timeline and Milestones

All project documents and final files are uploaded to the project GitHub repository. The four DEPI deadlines are fixed; all activities below are planned and none is reported as completed.

Phase	Main work	Deadline
1. Planning and Requirements	Proposal, roles, planning, risk plan and the 40 business questions	16 October 2026
2. Analysis and Data Design	Data dictionary, data model, cleaning plan and dashboard sketches	6 November 2026
3. Implementation	Data preparation, DAX measures and Power BI dashboard with filters	30 November 2026
4. Final Testing and Presentation	Validation, report, slides, demo video and final delivery	4 December 2026

7.1 Weekly plan

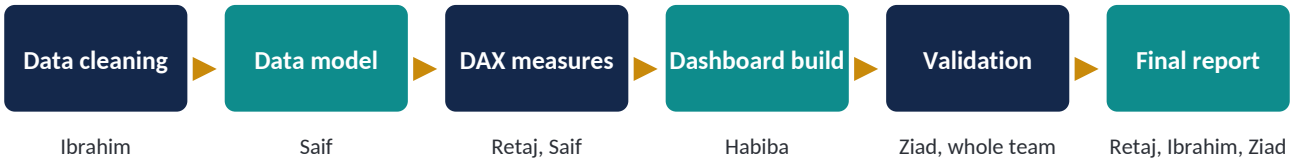
Dashed lines mark the four DEPI deadlines



Period	Planned work
16–23 October	Set up and organise the GitHub repository; confirm the 40 questions.
24 October – 6 November	Data dictionary, data model, cleaning plan and dashboard sketches.
7–20 November	Clean the data, build DAX measures and start the dashboard pages.
21–30 November	Finish the dashboard and check every number against the cleaned data.
1–4 December	Final report, demo video and final upload; slides are prepared earlier, in parallel.

7.2 Dependencies

Each stage depends on the one before it: measures need a stable cleaned dataset and model, the dashboard needs reviewed measures, validation needs the finished dashboard, and the final report uses only validated results.



Slides and the demo video are prepared in parallel with the dashboard build.

8 Dashboard Structure and Key Performance Indicators

8.1 Dashboard pages

The Power BI dashboard is planned as six pages. Page content below is intended design, not a completed result.

Page	Intended content
1. Overview	KPI cards, severity distribution, accident records by weekday and by hour.
2. Driver	Driver age band, sex, driving experience, education level, driver-vehicle relation and licence status, each with severity comparison.
3. Vehicle	Vehicle type, owner type, service years, recorded defects, vehicle movement and number of vehicles involved.
4. Road and Junction	Junction type, lanes or medians, alignment, surface type, area type and collision type.
5. Environment and Time	Hour and day patterns, weekend share, lighting, weather and road-surface condition.
6. Causes and Casualties	Reported causes and top-five causes, casualty counts, casualty class, age band and severity distribution, with “na” shown explicitly.

Slicers: accident severity, day of week, hour, weather, light condition, vehicle type and driver age band. Only the filters relevant to each page are shown; reported cause, collision type, driving experience and road-surface condition are recommended additions where they help the page.

8.2 Key performance indicators

KPI card	What it shows
Total Accidents	All accident records in the current filter (12,316 before any filter).
Serious or Fatal Share	Serious Injury plus Fatal injury records divided by all records in the current filter.
Fatal Accidents	Fatal injury records (158 before any filter).
Average Casualties	Average number of casualties per accident record.
Average Vehicles Involved	Average number of vehicles involved per accident record.
Top Cause	Most frequent reported cause in the current filter.
Peak Hour	Hour with the most accident records in the current filter.

8.3 Planned DAX measures

Total Accidents, Total Casualties, Total Vehicles Involved, Serious or Fatal Accidents, Fatal Accidents, Serious Share %, Fatal Share, Share of Total %, Average Casualties, Average Vehicles Involved and ranking (Top N) measures. Rates always divide by the accidents in the current filter.

8.4 Design principles

- Show counts and percentages together, with group sizes, so readers can compare severity across categories.
- Flag or hide groups with fewer than 30 accidents as too small to compare.
- Show a short data-limitations note and display “na” and blank values explicitly.
- Keep a simple, consistent layout with clear titles, cards and slicers; page load and filter response should stay under 3 seconds.
- Link every recommendation to a displayed finding and a stated limitation.

9 Risk Assessment and Mitigation

Likelihood and impact are rated Low, Medium or High. The main risks are small groups, missing data and wrong interpretation of results.

Risk	Likelihood	Impact	Mitigation
Missing or inconsistent values (up to 36% blank in one column)	High	Medium	Profile columns first; document decisions; keep “Unknown”, “na” and blanks separate; do not guess values.
Small groups give unstable rates (158 fatal records, rare vehicle types)	High	High	Show counts next to rates; flag groups with fewer than 30 accidents; avoid strong claims.
Association read as cause	Medium	High	Use wording such as “associated with”; never claim a factor caused an accident.
Casualty data handled incorrectly (grain, “na”, casualty columns used as risk factors)	Medium	High	Confirm what one row represents before summing; keep casualty columns for description only.
Incorrect DAX or dashboard totals that do not match the cleaned data	Medium	High	Compare key totals with Excel pivot tables; second-member review of every measure.
Percentages and Top N visuals break under filters	Medium	High	Test every slicer; build rates to divide by the accidents in the current filter.
Limited data: no date, location, speed or driver age in years	Certain	Medium	State the scope clearly; do not promise maps, trends or speed analysis.
Power BI file lost or overwritten	Medium	High	Upload a version to GitHub after each phase; one member edits the model at a time.
CSV path tied to one computer, so refresh fails for others	High	Medium	Use a file-path parameter or keep the CSV inside the repository.
Deadline pressure and tool-learning constraints	Medium	High	Prepare slides and demo video in parallel; assign tasks clearly; keep to the agreed tool plan.
A team member is late with a task	Medium	High	Weekly progress check by the Team Leader; members help each other.

10 Project Success Criteria and Deliverables

10.1 Success criteria

Project KPI	Target	How it is checked
Business questions addressed in the dashboard	40 of 40 (where the data supports them)	Tick each question against a chart, card or table
Dashboard totals reconciled with the cleaned data	100%	Compare cards with Excel pivot tables
DAX measures peer-reviewed	All measures	A second member reviews each measure
Dashboard pages, filters and navigation tested	All pages	Test every slicer and page before delivery
Source columns documented in the data dictionary	32 of 32	Compare against the source file
Deliverables organised in GitHub	All, before 4 December 2026	Team Leader checks the repository
Phase deadlines met	4 of 4	Weekly progress check
Final report, slides and demo video prepared	All three	Review before final delivery

10.2 Deliverables

Deliverable	Description	Owner
Cleaned dataset and cleaning log	Power BI-ready data; every change with reason and effect	Ibrahim
Data dictionary	All 32 source columns with meaning and data type	Saif
Data model	Schema diagram and checked relationships	Saif
DAX measures	Required measures, peer-reviewed	Retaj, Saif
Power BI dashboard	Six pages with charts, filters and KPIs	Habiba
Final report	Key findings, limitations and recommendations	Retaj, with Ibrahim and Ziad
Presentation slides	Final presentation	Ziad
Demo video	1–2 minute video of the dashboard, filters and main findings	One member, to be confirmed
GitHub repository	Project files and documentation	Retaj; all members upload

11 Final Conclusion and Repository

Analytical value. Road Traffic Accident Analysis turns 12,316 accident records into a structured, reproducible analysis. By cleaning the data carefully, keeping missing values visible and answering 40 defined questions in one Power BI dashboard, the project will make it easier to see when accidents are recorded and which driver, vehicle, road, environment and cause conditions are associated with serious and fatal outcomes.

Road-safety contribution. The intended contribution is evidence for discussion: clear comparisons with counts and group sizes that support safety recommendations linked to displayed findings.

Limitations. The dataset has no date, location or speed, a large share of values is missing in some columns, and only 158 records are fatal. Results are descriptive associations, not proof of causation, and are not suitable for real-time or location-specific decisions.

Next steps. Confirm the 40 questions and the GitHub structure, complete the data dictionary, model and cleaning plan by 6 November 2026, implement cleaning, DAX measures and the dashboard by 30 November 2026, and complete validation, the report, slides and demo video by 4 December 2026.

GITHUB REPOSITORY

https://github.com/retajelgallad/Road-Accidents-Analysis